

Evolutionary Robotics

1: introduction

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**Hochschule
Bonn-Rhein-Sieg**
University of Applied Sciences

Outline

Introduction

Biological fundamentals

Evolutionary algorithms

Artificial neural networks

How to evolve robots

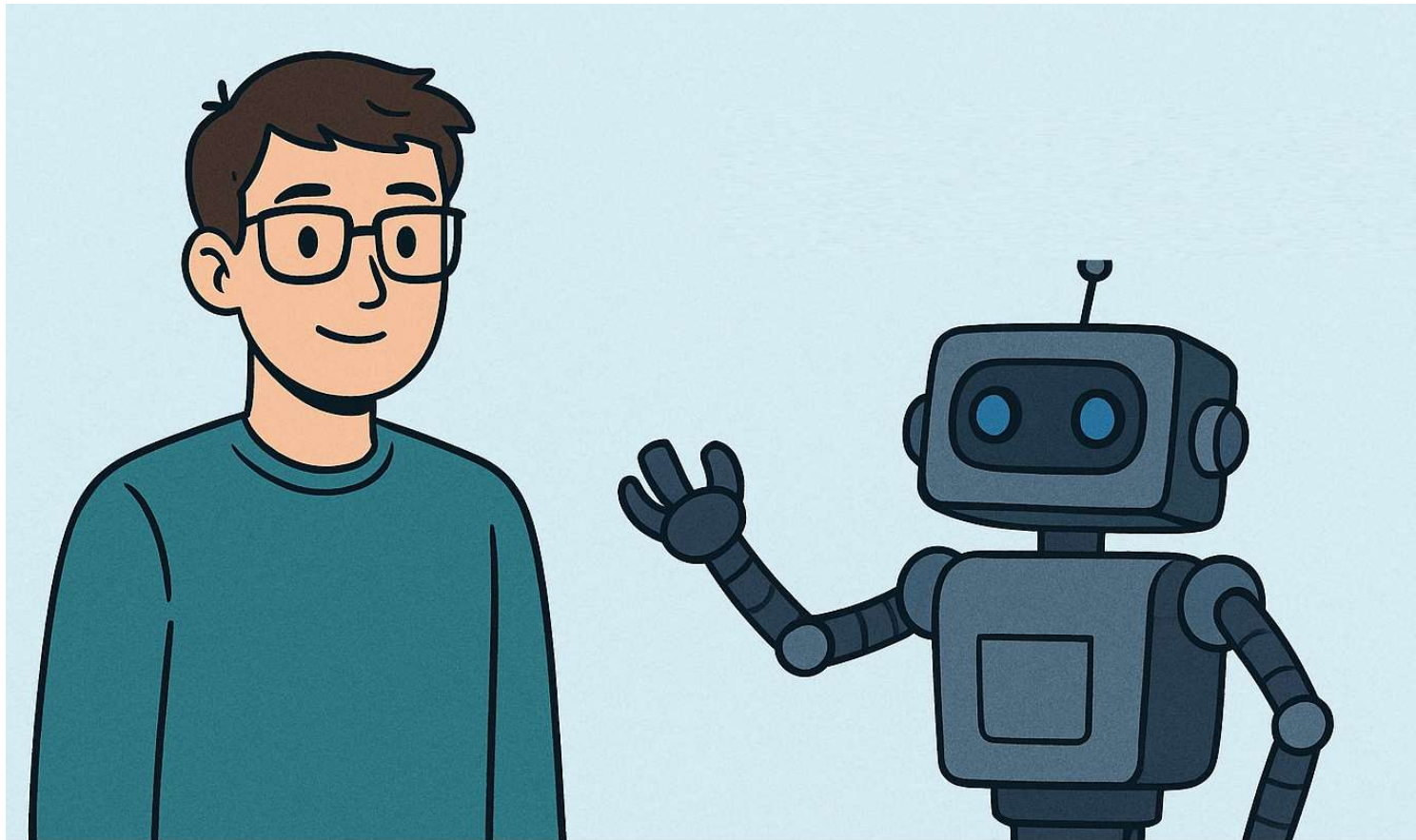
Evolution of simple navigation

Power and limits of reactive intelligence

Evolving also morphology

Challenges and new approaches

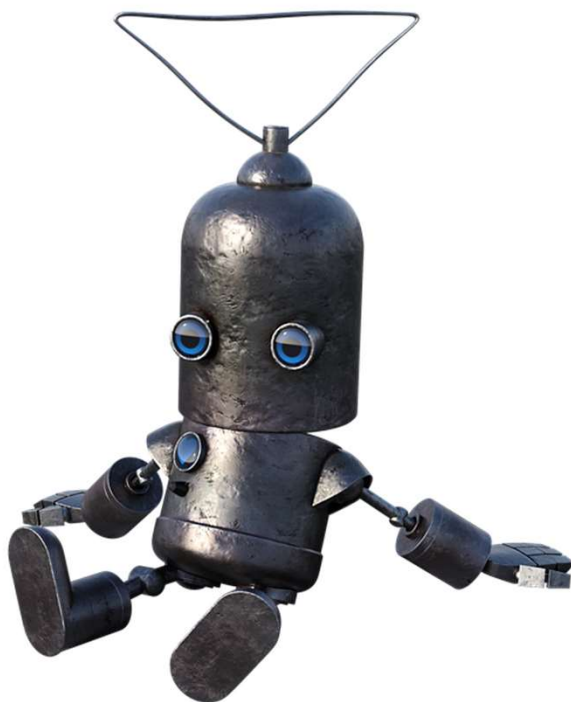
Introduce yourself





Course Preliminaries

What is this Course About?



roughly there are 3 ways of programming a robot

- by hand
- by machine learning
- by artificial evolution

main components we'll study

- natural evolution
- evolutionary computation
- artificial neural networks
- robot simulations
- the art of evolving useful behaviors quickly that can actually be used on real robots

Course Formalities

15 lectures + lab classes

- Lectures every **Thursday** between 13:30-15:00 in **C120**
- Lab classes every **Thursday** between 15:15–16:45 in **C120**
 - We will have occasional **homework** assignments about the course material

grading based on **single oral final exam**,

+ required min. 30% of each task sheet

+ required min. 50% total

(please submit in groups of 2)

exam will have questions **about the course material**

Outline

Introduction
Why Evolutionary Robotics?

Biological fundamentals
(Natural Selection, neural Systems)

Evolutionary algorithms
(Selection, Mutation, Crossover)

Artificial neural networks
(Controller for robots)

How to evolve robots
(Encoding, Fitness, Simulation)

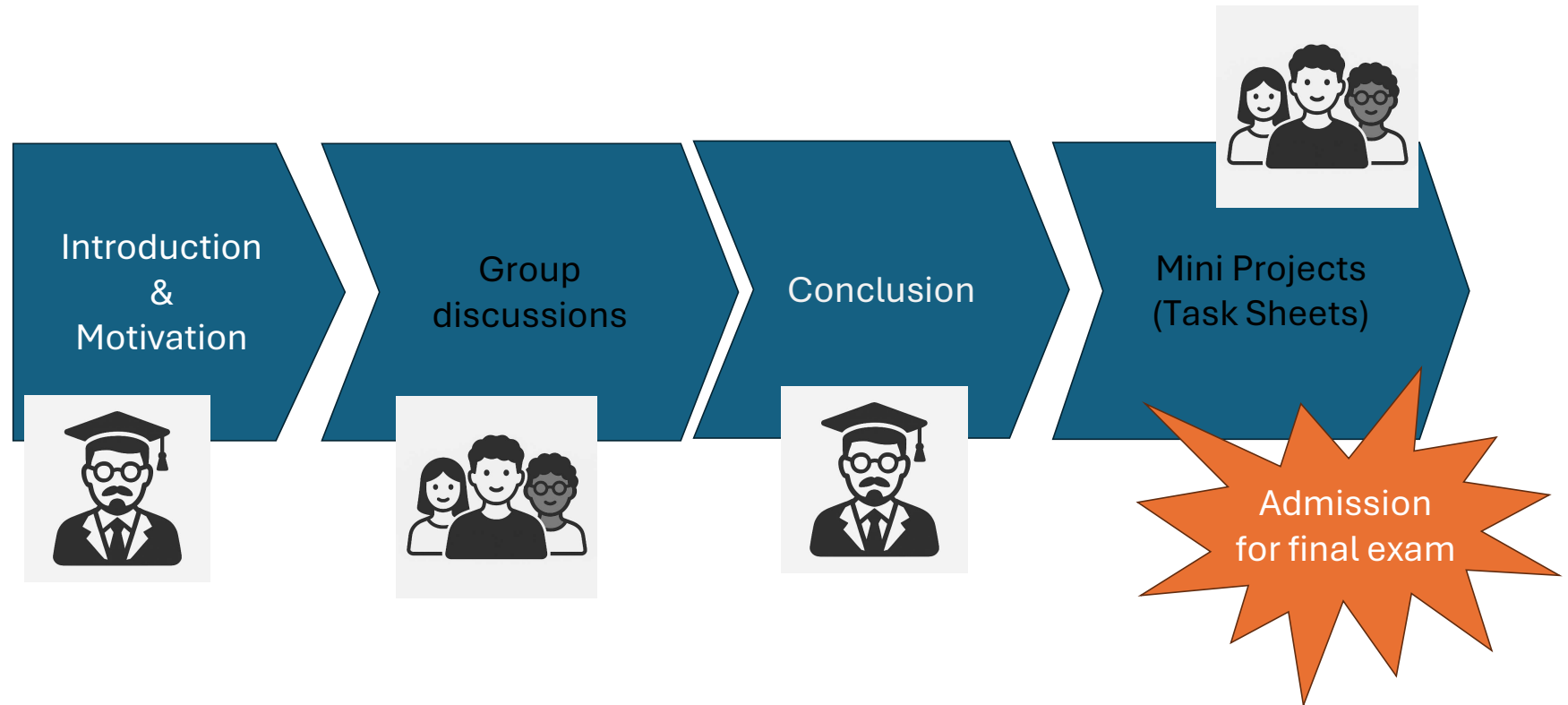
Evolution of simple navigation
(Phototaxis, Obstacle avoidance)

Power and Limitations of Reactive
Intelligence

Evolving Morphology
(Body + Brain Co-evolution)

Challenges and new approaches
(Reality Gap, Diversity, DNNs)

Teaching: Structure of every Session



My Teaching Philosophy

Each session ...

- begins with an introduction and motivation/problem statement
- Followed by group discussions to produce **concepts** and **solutions**
- Each group presents their solutions and ideas

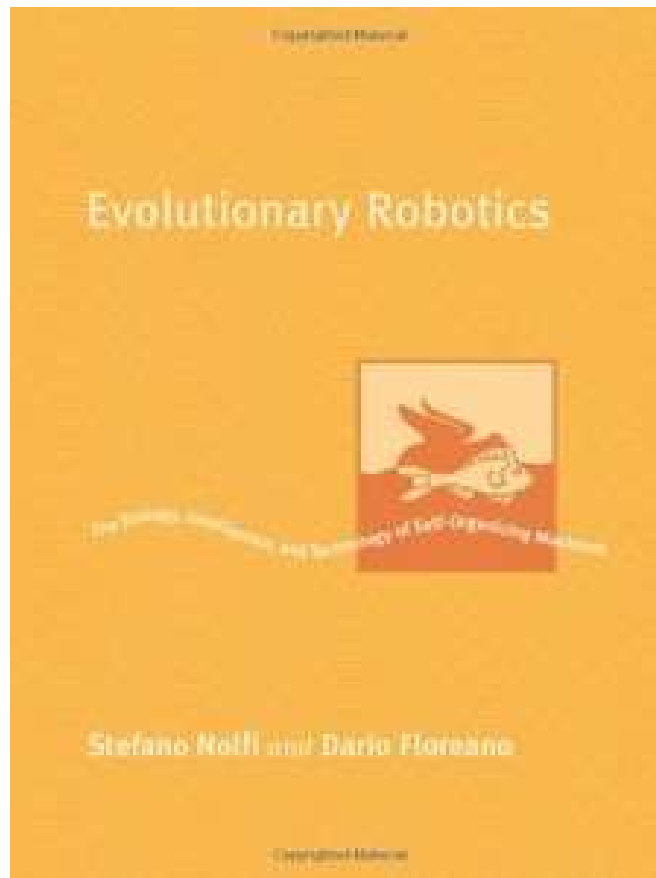
I will be on your side to improve your results

Slides + group discussion notes will be published on lea after each lecture

Exp. Skills Eng. Rob. Msc → Teaching Philosophy

- Research and Problem-Solving Skills
 - Capability to conduct independent research and critically evaluate scientific literature.
 - Systematic problem-solving and innovation in robotic applications.
- Project Management and Collaboration
 - Ability to manage projects, write technical documentation, and publish research.
 - Collaboration skills for interdisciplinary teams (mechanical, electrical, software).
- Ethics and Safety Awareness
 - Understanding of safety standards, regulations, and ethical considerations in robotics.
- Practical Experience
 - Hands-on experience with prototyping, simulation, and real-world deployment.
 - Familiarity with tools like Gazebo, V-REP, or Webots for virtual testing.
- Motivation
- Group Discussions
- Solution

Bibliography



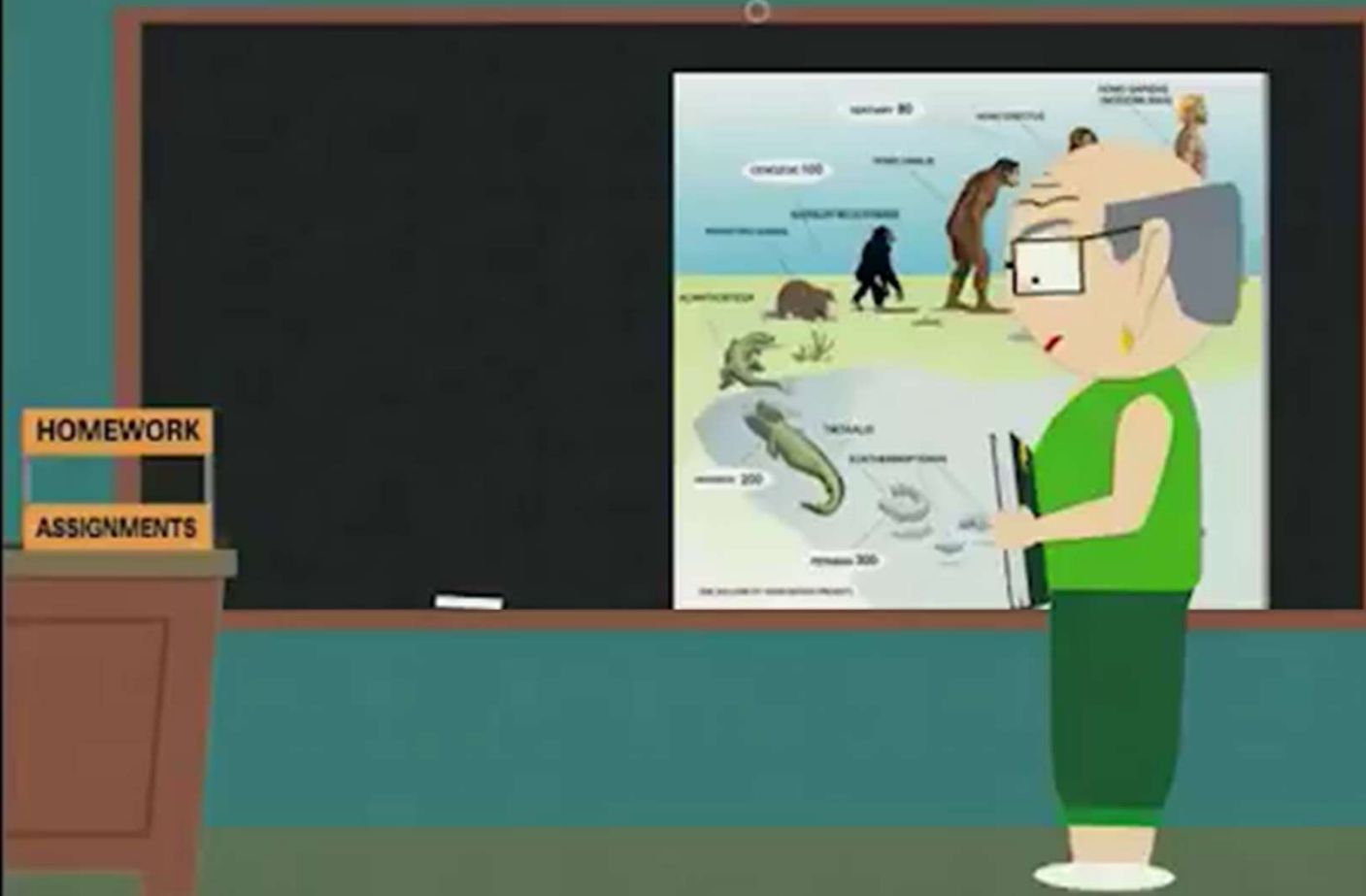
Recommended book

*Evolutionary Robotics:
The Biology, Intelligence, and Technology of Self-Organizing
Machines*

by Stefano Nolfi and Dario Floreano
MIT Press, 2004

(about 25+ EUR)

Aa Bb Cc Dd Ee Ff Gg Hh Ii Jj Kk Ll Mm Nn Oo Pp Qq Rr Ss Tt Uu Vv Ww Xx Yy Zz



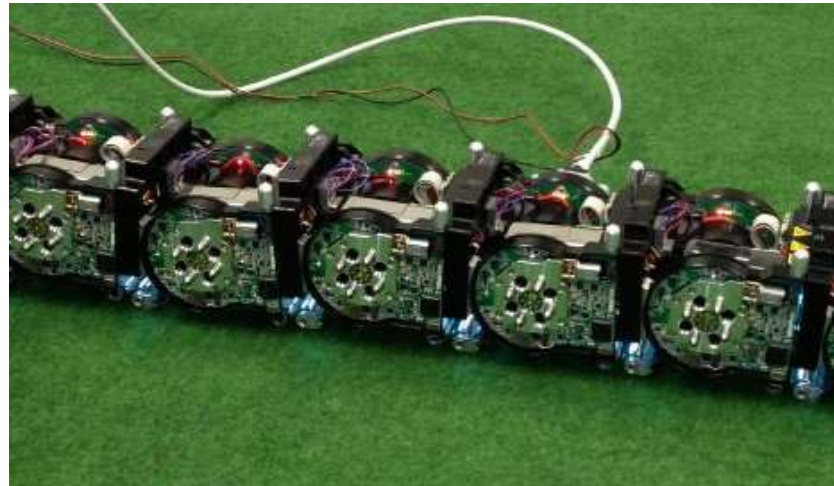
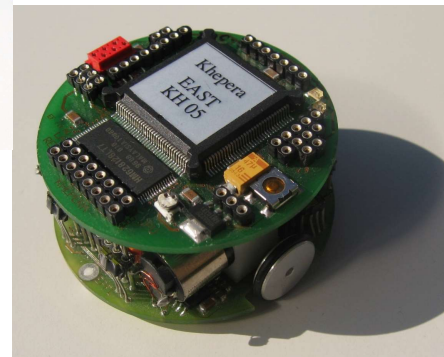
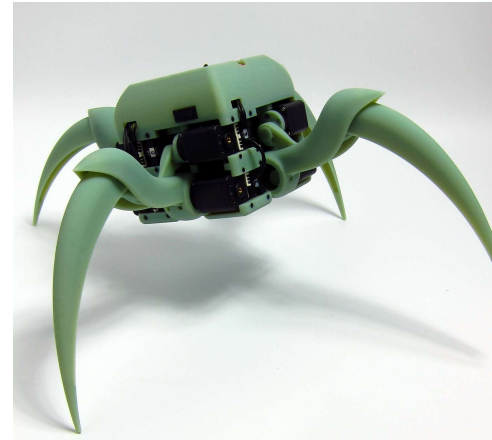
What is evolutionary robotics?

Evolution

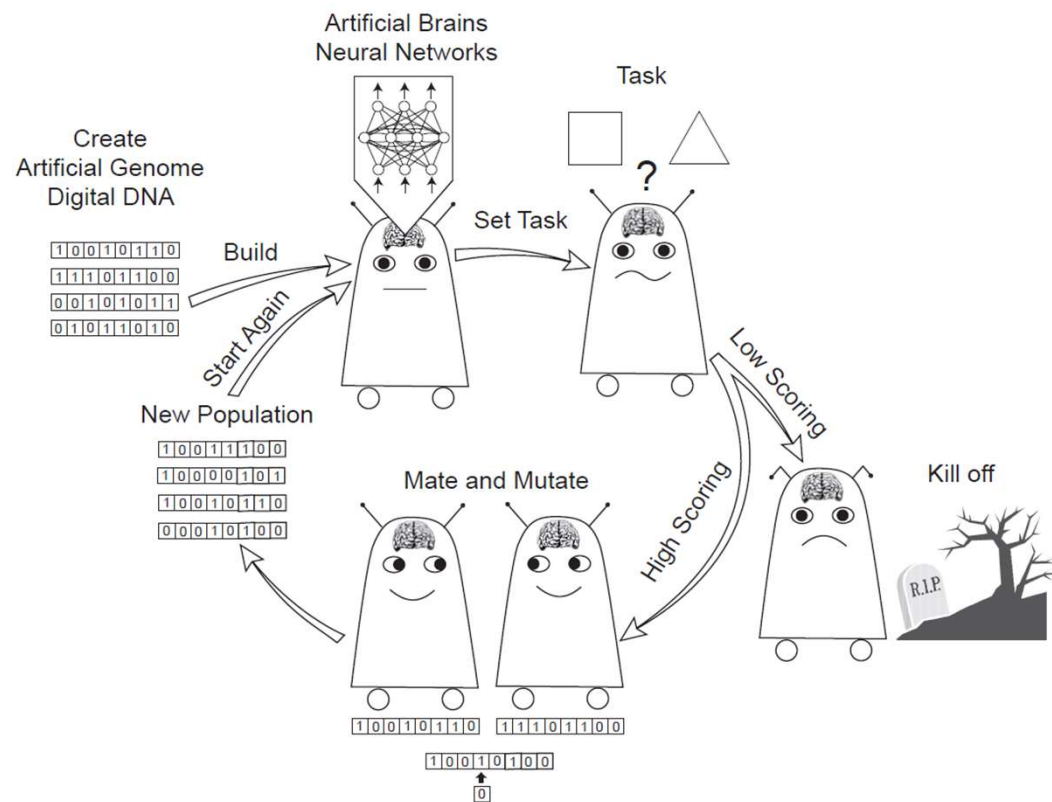
Evolutionary computation

Machine learning

Robotics (autonomy, behavior)



Evolutionary Robotics



Evolutionary Robotics

Communications of the ACM, Volume
56, Issue 8, August 2013,

Pages 74-83

<http://doi.acm.org/10.1145/2493883>

DOI:10.1145/2493883

**Taking a biologically inspired approach to
the design of autonomous, adaptive machines.**

BY JOSH C. BONGARD

Evolutionary Robotics

THE AUTOMATED DESIGN, construction, and deployment of autonomous and adaptive machines is an open problem. Industrial robots are an example of autonomous yet nonadaptive machines: they execute the same sequence of actions repeatedly. Conversely, unmanned drones are an example of adaptive yet non-autonomous machines: they exhibit the adaptive capabilities of their remote human operators. To date, the only force known to be capable of producing fully autonomous as well as adaptive machines is biological evolution. In the field of evolutionary robotics,⁹ one class of population-based metaheuristics—evolutionary algorithms—are used to optimize some or all aspects of an autonomous robot. The use of metaheuristics sets this subfield of robotics apart from the mainstream

about the problem: evolutionary algorithms can improve both the parameters and the architecture of the robot's control policy, and even the shape of the robot itself.

Because the trial-and-error nature of evolutionary algorithms requires a large number of evaluations during optimization, in many evolutionary robotics experiments optimization is first carried out in simulation. Typically an evolutionary algorithm generates populations of virtual robots that behave within a physics-based simulation.^b Each robot is then assigned a fitness value based on the quality of its behavior. Robots with low fitness are deleted while the robots that remain are copied and slightly modified in some random manner. The new robots are evaluated in the simulator and assigned a fitness, and this cycle is repeated until some predetermined time period has elapsed. The most-fit robot may then be manufactured as a physical machine and deployed to perform its evolved behavior.

To illustrate the distinction between mainstream and evolutionary robotics, consider two experiments drawn from the two fields. Legged locomotion—

^b Interested readers may download and perform their own evolutionary robotics experiments at <http://www.uvm.edu/~ludobots>.

» key insights

- Manual design of a mobile robot that is autonomous and adaptive is extremely difficult.

From evolutionary computation to the evolution of things

Agoston E. Eiben¹ & Jim Smith²

Evolution has provided a source of inspiration for algorithm designers since the birth of computers. The resulting field, evolutionary computation, has been successful in solving engineering tasks ranging in outlook from the molecular to the astronomical. Today, the field is entering a new phase as evolutionary algorithms that take place in hardware are developed, opening up new avenues towards autonomous machines that can adapt to their environment. We discuss how evolutionary computation compares with natural evolution and what its benefits are relative to other computing approaches, and we introduce the emerging area of artificial evolution in physical systems.

The proposal that evolution could be used as a metaphor for problem solving came with the invention of the computer¹. In the 1970s and 1980s the principal idea was developed into different algorithmic implementations under names such as evolutionary programming², evolution strategies^{3,4} and genetic algorithms⁵, followed later by genetic programming⁶. These branches merged in the 1990s, and in the past 20 years so-called evolutionary computation or evolutionary computing has proven to be highly successful across a wide

evolutionary algorithm one has to go through a number of design steps. The first step is to identify a representation: a suitable data structure that can represent possible solutions to the problem. The next step is to define a way of measuring the quality of an individual based on problem-specific requirements. The final step is to specify suitable selection and variation operators (Fig. 1).

Analogous to natural evolution, an evolutionary algorithm can be thought of as working on two levels. At the higher level (the original

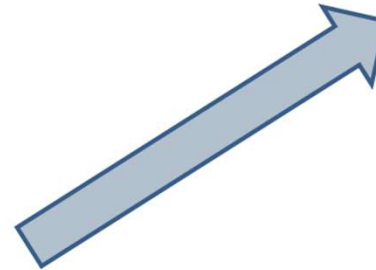
Nature 521, 476-482 (May 2015)

<http://dx.doi.org/10.1038/nature14544>

Important idea for Evol. Rob.

Embodied Cognition

Reactive vs. deliberative approach



Origins

1: Introduction

“Cogito ergo sum.”



“I think, therefore I am.”



“Do I exist?”



“There is something that is asking the question, Because there is an 'I' in the sentence. Whatever that 'I' thing is, it exists.”



The soul(/mind) surely exists; the body, I'm not so sure about.



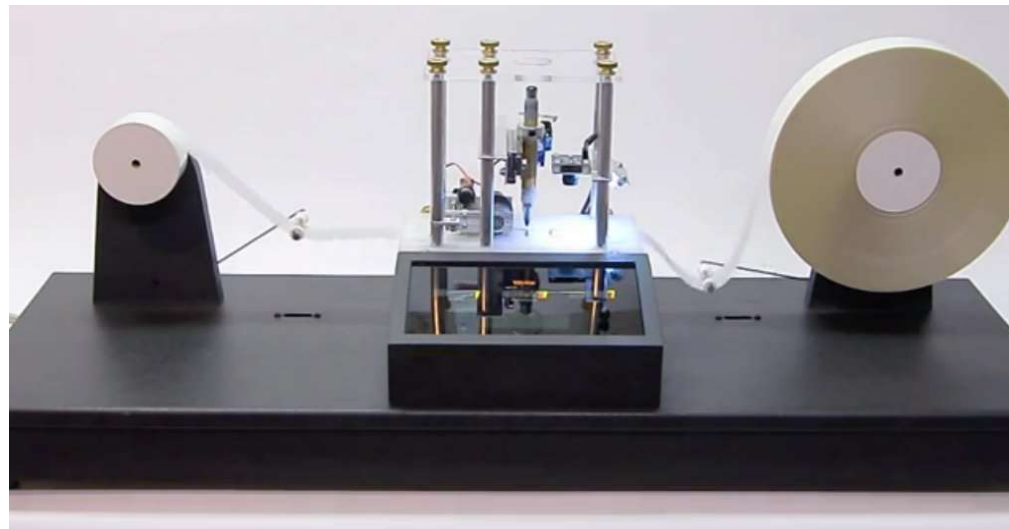
So the soul(/mind) and body are different.



Cartesian Dualism; “The Mind/Body Problem”

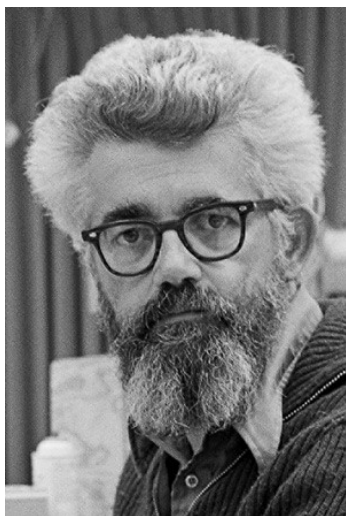
René Descartes
(1596-1650)

Traditions of computer science



Origins of Artificial Intelligence

The Dartmouth Summer Research Conference on Artificial Intelligence (1956)
(Dartmouth College, New Hampshire, USA)



John
McCarthy



Marvin
Minsky



Nathaniel
Rochester



Claude
Shannon

Artificial Intelligence (1/3)

The Dartmouth Summer Research Conference on Artificial Intelligence (1956)
(origin of the word 'artificial intelligence')

Proposal: “We propose that a 2 month, 10 man (sic) study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to **simulate** it.

An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and **improve themselves**.

We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists works on it together for a summer.”

Artificial Intelligence (2/3)

“The first period of excitement, which began with the Dartmouth meeting, was later described by John McCarthy (the event’s main organizer) as the ‘Look, Ma, **no hands!**’ era. During these early days, researchers built systems designed to refute claims of the form ‘**No machine could ever do X!**’ ”

– Superintelligence, Bostrom

“Expert **chess** playing, for example, was once thought to epitomize human intellection. In the view of several experts in the late fifties: ‘If one could devise a successful chess machine, one would seem to have penetrated to the core of human intellectual endeavor.’ This no longer seems so. One sympathizes with John McCarthy, who lamented: ‘**As soon as it works, no one calls it AI anymore.**’ ”

– Superintelligence, Bostrom

Artificial Intelligence (3/3)

“A follow-up program, the **General Problem Solver**, could in principle solve a wide range of formally specified problems. Programs that could solve calculus problems typical of first-year college courses, visual analogy problems of the type that appear in some IQ tests, and simple verbal algebra problems were also written. The **Shakey robot** (so named because of its tendency to tremble during operation) demonstrated how logical reasoning could be integrated with perception and used to plan and control physical activity. The **ELIZA** program showed how a computer could impersonate a Rogerian psychotherapist. In the mid-seventies, the program **SHRDLU** showed how a simulated robotic arm in a simulated world of geometric blocks could follow instructions and answer questions in English that were typed in by a user.”

– Superintelligence, Bostrom

ELIZA

An example of classical Artificial Intelligence

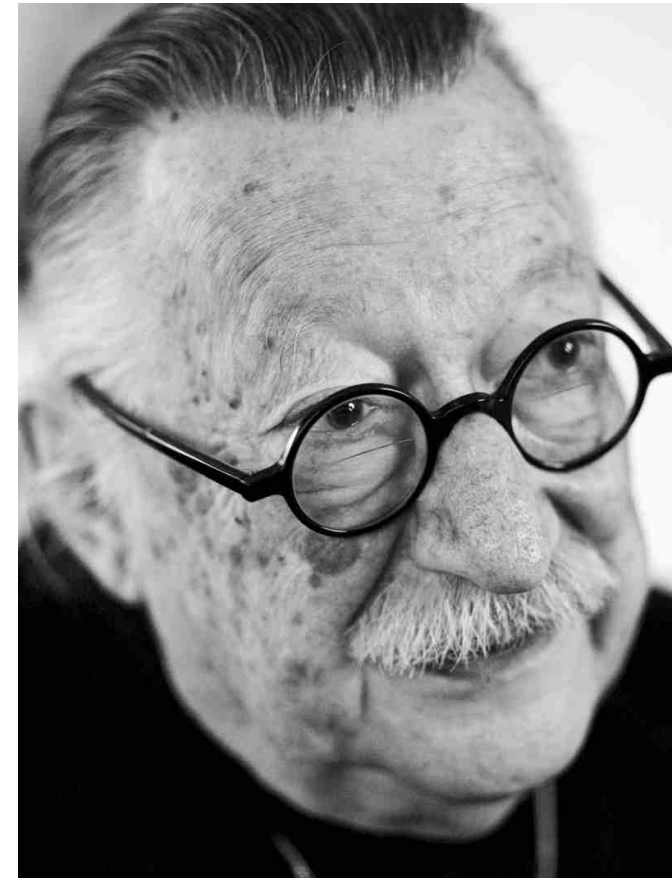
Created by Weizenbaum between 1964 to 1966

⇒ Internet chatbots

Sample ELIZA Rule:

User: “I [verb] [proper noun]”

ELIZA: “Why do you [verb] [proper noun]?”



Joseph Weizenbaum
(1923-2008)

“The computer scientist Donald Knuth was struck that ‘AI has by now succeeded in doing essentially everything that **requires ‘thinking’** but has failed to do most of what people and animals do **‘without thinking’**—that, somehow, is much harder!’ Analyzing visual scenes, recognizing objects, or **controlling a robot’s behavior** as it interacts with a natural environment has proved challenging.”

– Superintelligence, Bostrom

A New Paradigm

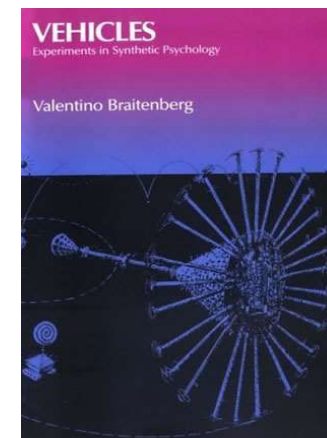
1: Introduction

Valentino Braitenberg, Vehicles

A series of (hypothetical) simple robots

Seem, to the outside observer, to exhibit complex behavior

Not from a complex brain, but from
a simple agent interacting with a rich environment.



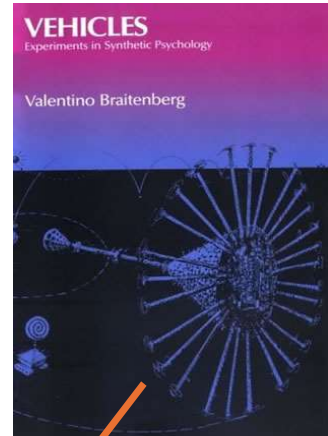
[https://mitpress.mit.edu/
books/vehicles](https://mitpress.mit.edu/books/vehicles)

Valentino Braitenberg, Vehicles

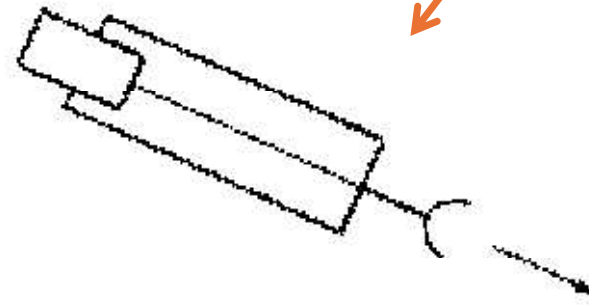
Vehicle 1 – Getting around:

A single sensor is attached to a single motor. Propulsion of the motor is proportional to the signal detected by the sensor.

The vehicle will always move in a straight line, slowing down in the cold, speeding up in the warm.

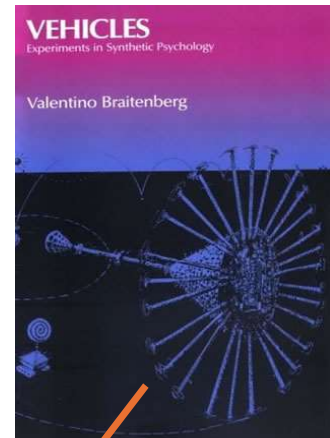


<https://mitpress.mit.edu/books/vehicles>

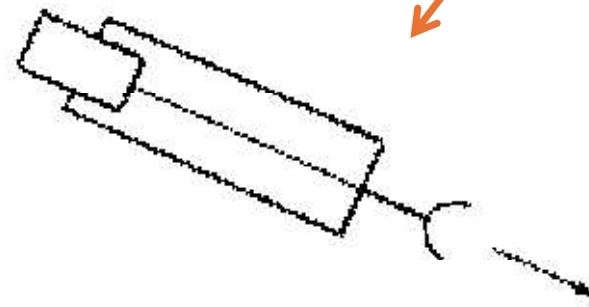


Valentino Braitenberg, Vehicles

Braitenberg: “Imagine, now, what you would think if you saw such a vehicle swimming around in a pond. It is restless, you would say, and **does not like warm water**. But it is quite stupid, since it is not able to turn back to the nice cold spot it overshot in its restlessness. Anyway, you would say, it is **alive**, since you have never seen a particle of dead matter move around quite like that.”



<https://mitpress.mit.edu/books/vehicles>

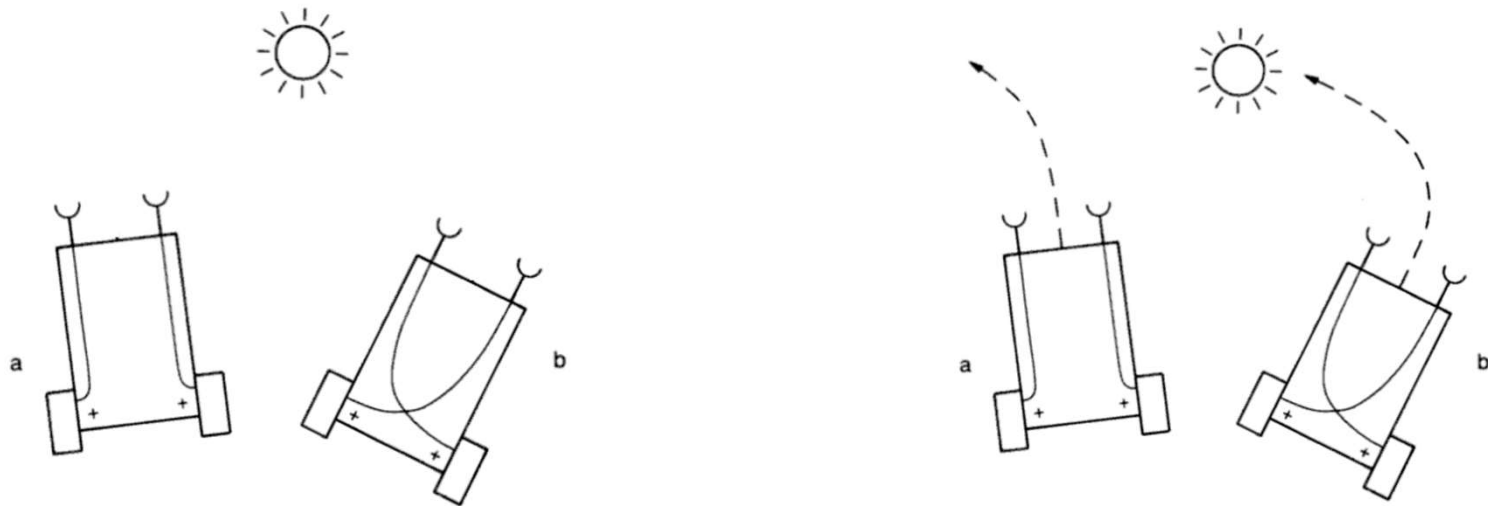


Braitenberg – Vehicles – 2

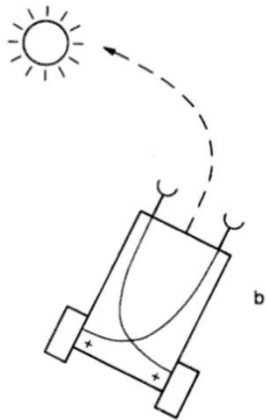
Vehicle series 2: Fear and aggression

Each vehicle has two sensors and two motors.

Still a directly proportionate relationship between the signal strength and force of the motor.

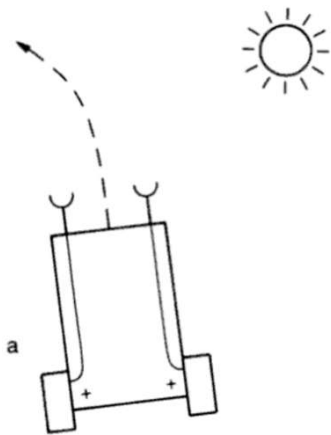


Braitenberg – Vehicles – 2



The Aggressor (b)

Left sensor to right motor; right sensor to left motor;
will charge the light if it's dead-ahead;
will charge the light if it's off to the side;
will eventually hit it, as long as the light stays in the vicinity.



The “Scaredy-Cat” (a)

Left sensor to left motor; right sensor to right motor;
will charge the light if it's dead-ahead;
will run from the light if it's off to the side;
will eventually stop at some distance with its back to the light.

Connectionism (1980s)

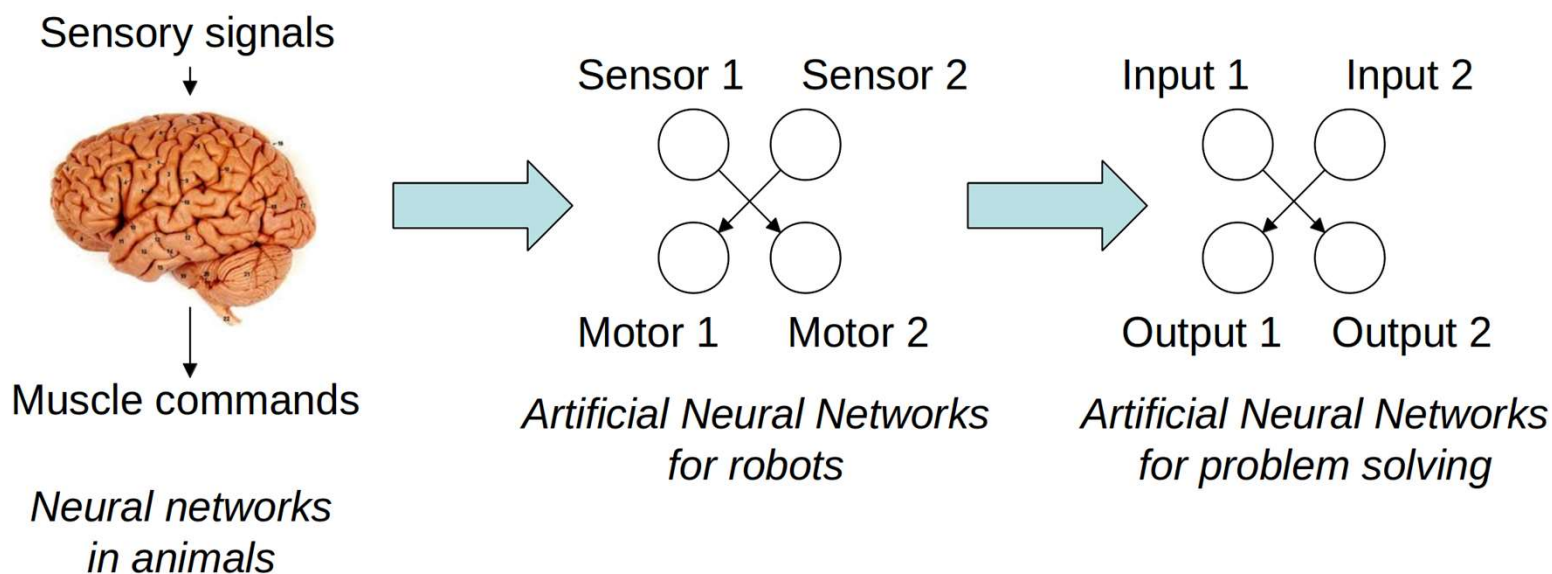
Sample ELIZA Rule:

IF User: “I [verb] [proper noun]”,

THEN: ELIZA: “Why do you [verb] [proper noun]?”

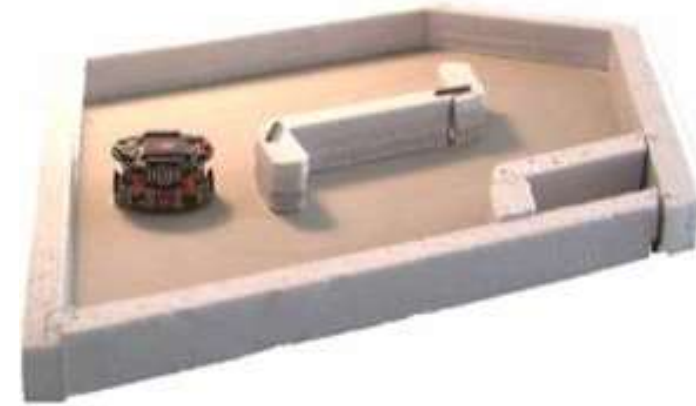
“mind-centered”

“body-centered”



Embodied cognition

A body is a tool for affecting the world,
and for being affected by it.



Embodied cognition:

The way you process information is affected by the fact that you have a body.

Example: If you have a body that can move, and you can see, then moving will cause immediate feedback.

Non-embodied technologies (such as computers) must wait for feedback from another computer (via packets) or a person (via an input device).

Example: Image Segmentation (1/2)

Challenge: How to recognize objects in a scene; where are the object's boundaries? This problem is known as “image segmentation” in the field of computer vision.

It is one of the most difficult problems in computer vision.



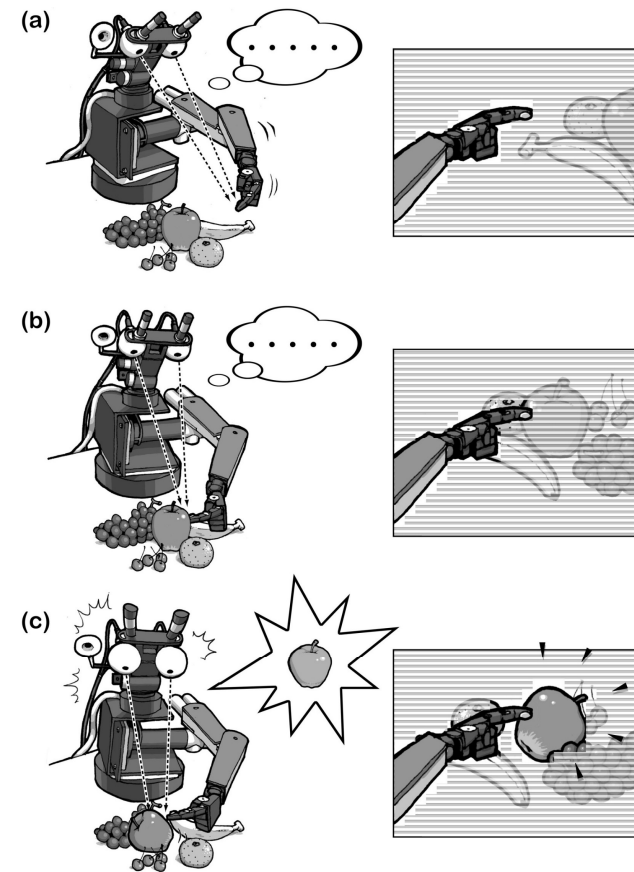
Example: Image Segmentation (2/2)

G. Metta, P. Fitzpatrick,
Early integration of vision and manipulation,
Adaptive Behavior, 2003
<http://dx.doi.org/10.1109/IJCNN.2003.1223994>

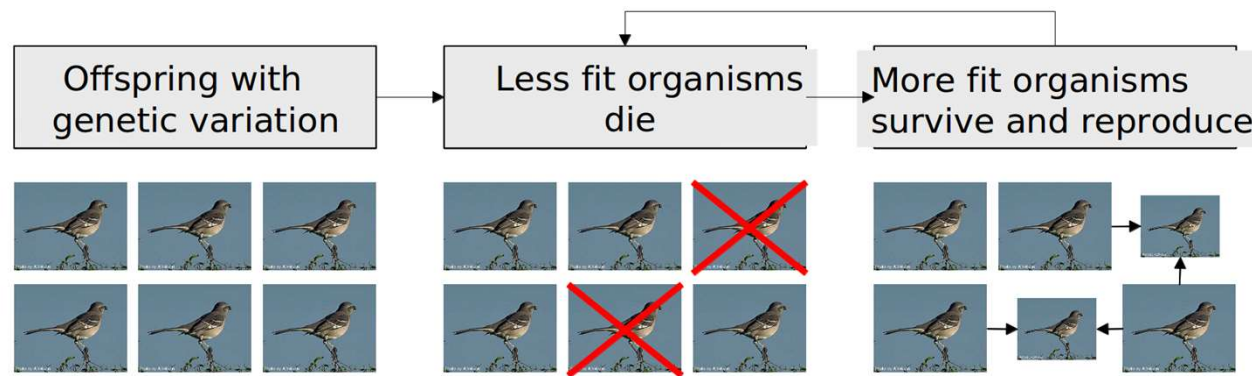
Challenge: How to recognize objects in a scene; where are the object's boundaries? This problem is known as "image segmentation" in the field of computer vision.

Solution:
Cog interacts with the scene, rather than passively observing it.

Q: What else can Cog infer about the objects it interacts with?

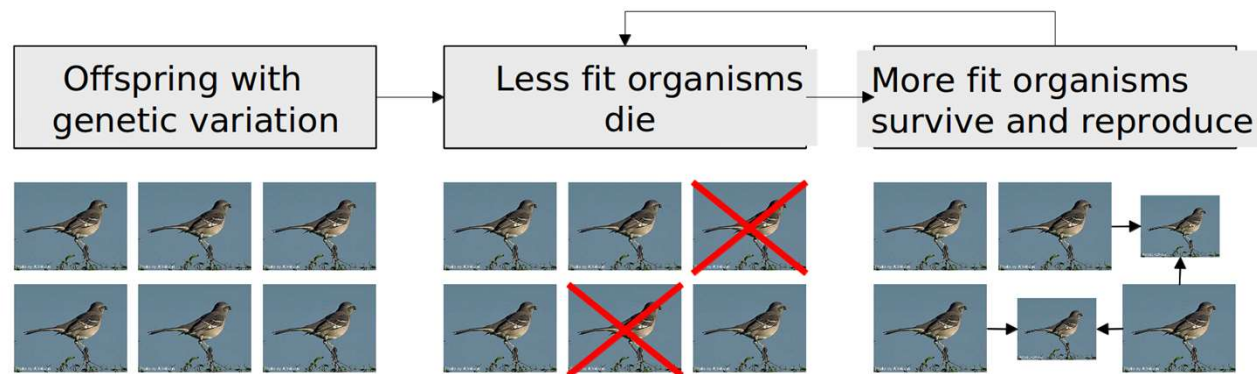


“Evolutionary Algorithms in 3 minutes”



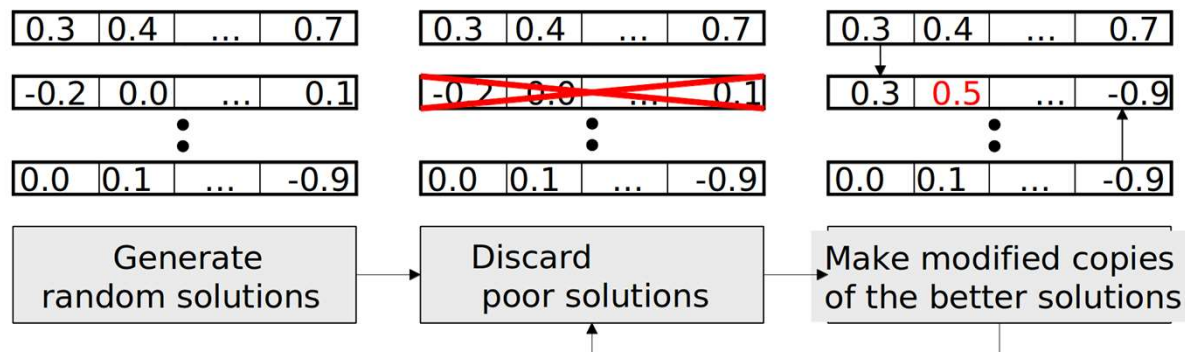
BIOLOGICAL EVOLUTION

“Evolutionary Algorithms in 3 minutes”

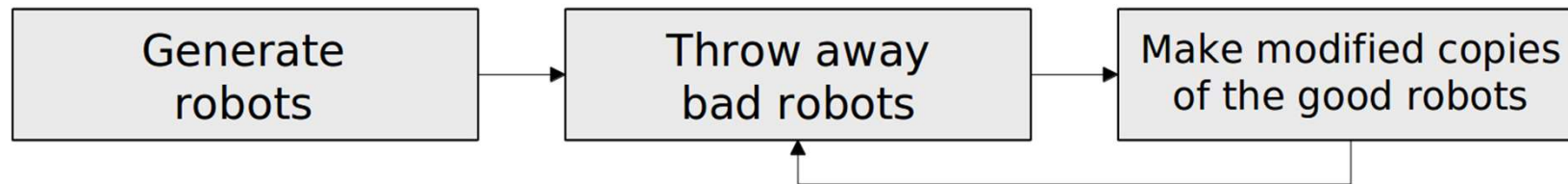


BIOLOGICAL EVOLUTION

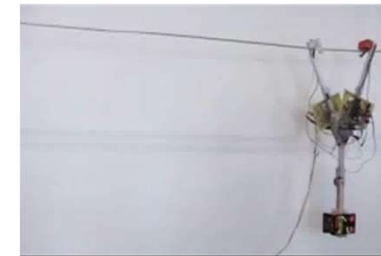
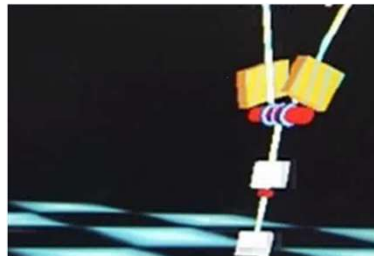
THE GENETIC ALGORITHM



Evolutionary Robotics

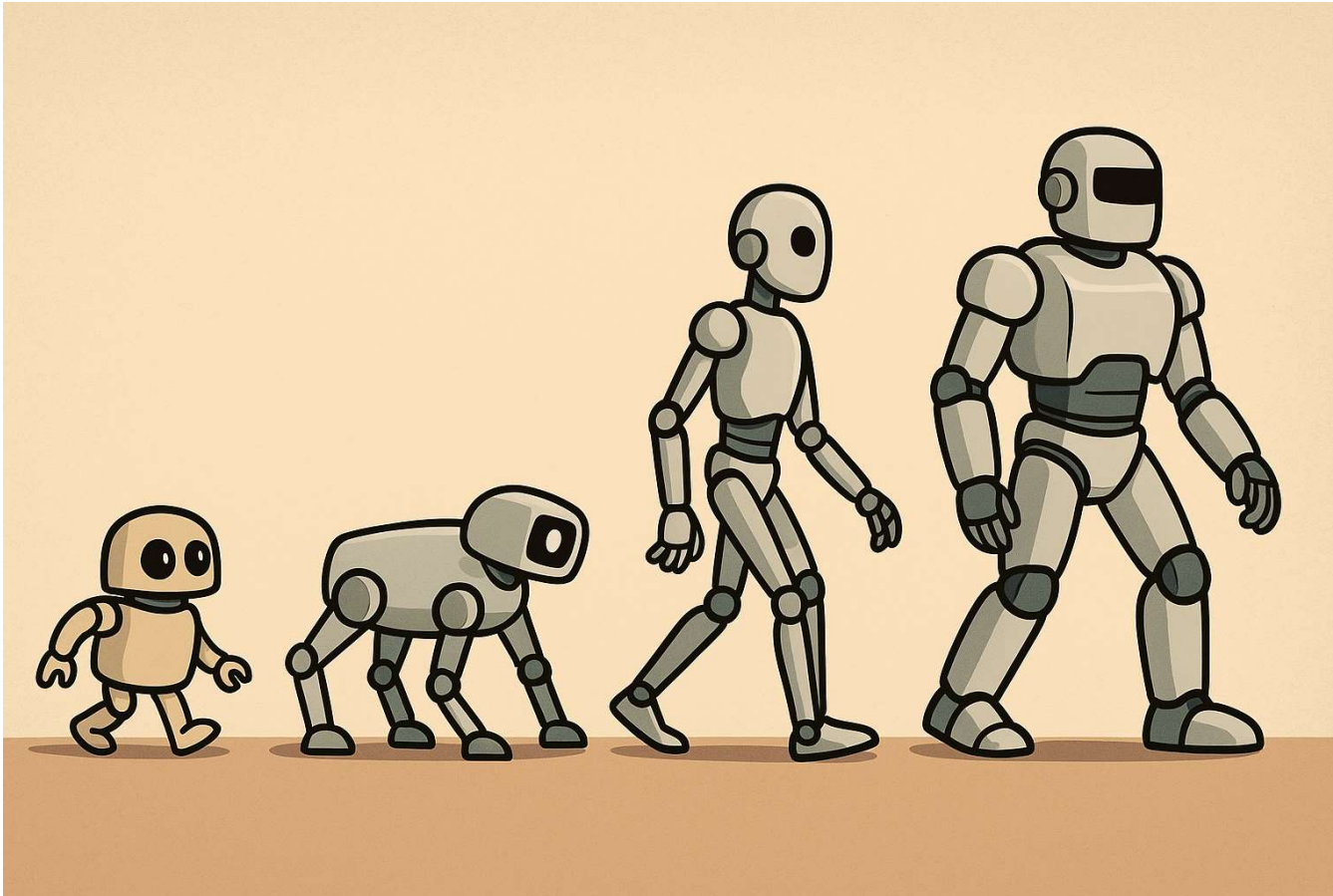


evolutionary algorithm + robot simulator = evolutionary robotics



Frutiger, D. R., J. C. Bongard and F. Iida (2002) "Iterative Product Engineering: Evolutionary Robot Design", in Bidaud, P. and F. B. Amar (eds.), Proceedings of the Fifth International Conference on Climbing and Walking Robots, Professional Engineering Publishing, pp. 619-629.

Evolution of Robots vs Evolutionary Robotics



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- Stefano Nolfi and Dario Floreano. *Evolutionary Robotics: The Biology, Intelligence, and Technology of Self-Organizing Machines*. MIT Press, 2000.

One Minutes of Thinking

Take one minute and think about this questions:
What is the highlight of the lecture today?

01:00

One Minutes of Thinking

Take one minute and write down the highlight in one sentence

01:00

One Minutes of Thinking

Take one minute and think about this questions:

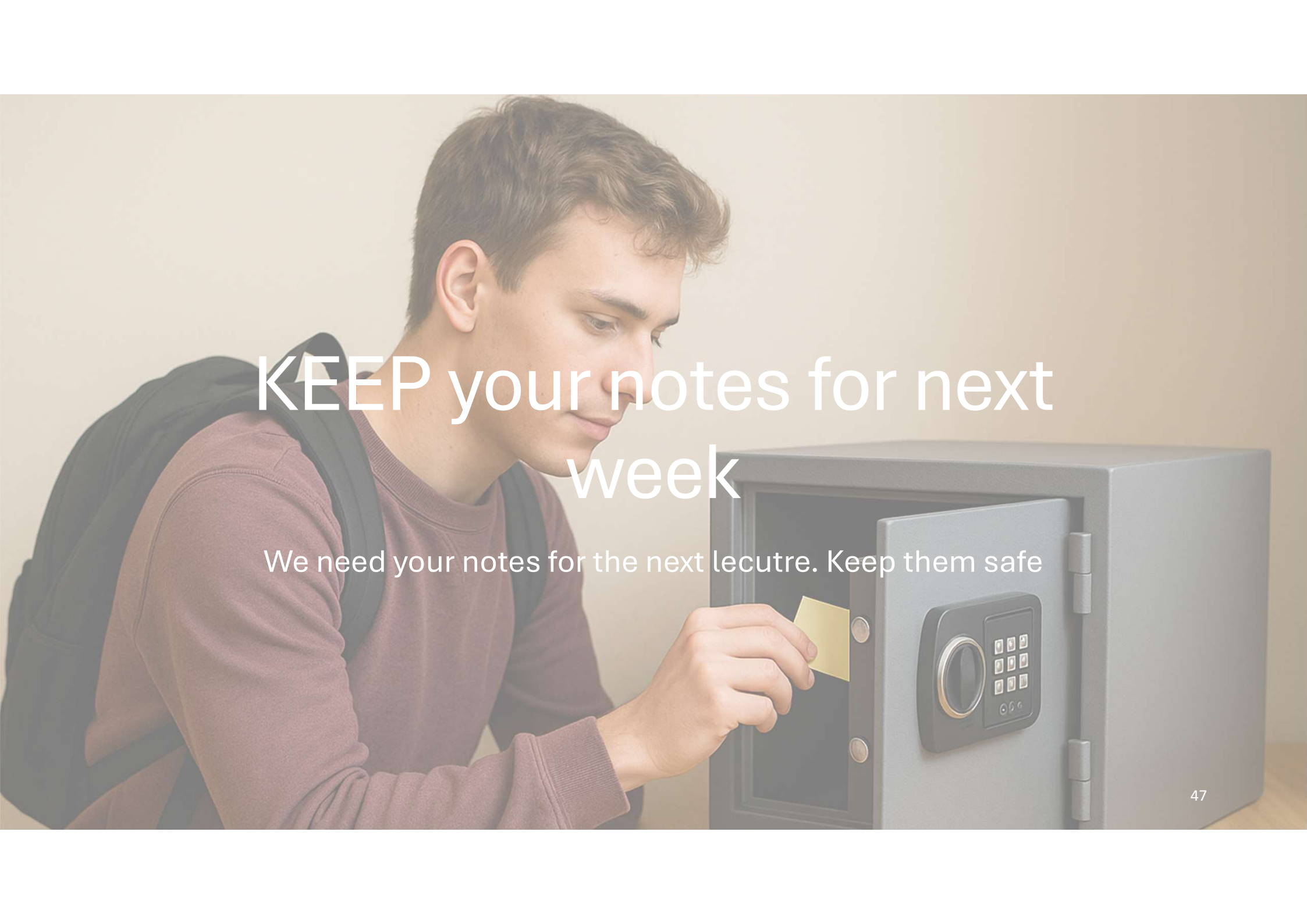
What is challenging for you to understand from the lecture today?

01:00

One Minutes of Thinking

Take one minute and write down the challenge in one sentence

01:00

A young man with brown hair, wearing a maroon sweater and a black backpack, is leaning over an open grey safe. He is holding a yellow sticky note and placing it inside the safe. The safe has a digital keypad and a handle. The background is a plain, light-colored wall.

KEEP your notes for next week

We need your notes for the next lecture. Keep them safe